

VILO ACUNA INTL, Cuba

WMO: 783334

Lat: 21.616N

Lon: 81.546W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (CUB)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 1             | 65.9       | 67.7 | 51.6                          | 56.8 | 68.7 | 53.9 | 61.7 | 70.2 | 24.2                  | 72.4 | 23.0 | 73.6 | 13.0                  | 20   | N/A |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.6                    | 90.0            | 78.6 | 89.8 | 78.4 | 89.5 | 78.2 | 81.2                | 86.7 | 80.6 | 86.4 | 80.0 | 86.0 | 9.4                  | 100  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.2                            | 151.7 | 84.7 | 79.0 | 150.6 | 84.8 | 78.7 | 148.8 | 84.8 | 45.0          | 87.9 | 44.4 | 86.7 | 43.8 | 86.3 | 84.6              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
| Min               |      | Max  |                            | Min  | Max  | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 22.0              | 20.2 | 18.6 |                            | DB   | 62.2 | 91.9               | 3.3                                                | 1.6       | 59.8 | 93.0       | 57.8 | 93.9       | 56.0 | 94.8       | 53.6 | 96.0 |
|                   |      |      |                            | WB   | 55.3 | 82.3               | 3.0                                                | 1.6       | 53.1 | 83.4       | 51.4 | 84.4       | 49.7 | 85.2       | 47.6 | 86.4 |

Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 80.7        | 75.4  | 76.7  | 77.4  | 80.3  | 81.7  | 83.7  | 84.7  | 84.8  | 84.4  | 82.4  | 79.1  | 77.3  |
|                                                                        | DBStd       | 4.11        | 3.31  | 3.02  | 2.69  | 1.67  | 2.04  | 1.88  | 1.75  | 2.02  | 2.03  | 2.37  | 2.66  | 3.70  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 1           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     |
|                                                                        | CDD50       | 11195       | 789   | 748   | 850   | 908   | 982   | 1010  | 1075  | 1078  | 1031  | 1005  | 874   | 845   |
|                                                                        | CDD65       | 5721        | 324   | 328   | 385   | 458   | 517   | 560   | 610   | 613   | 581   | 540   | 424   | 381   |
|                                                                        | CDH74       | 60079       | 1815  | 2173  | 2742  | 4392  | 5649  | 6972  | 8062  | 8159  | 7538  | 6179  | 3602  | 2796  |
|                                                                        | CDH80       | 20006       | 156   | 245   | 384   | 949   | 1641  | 2782  | 3665  | 3787  | 3310  | 2094  | 642   | 351   |
| (14) Wind                                                              | WSAvg       | 10.0        | 11.1  | 10.6  | 11.1  | 10.1  | 9.9   | 9.4   | 8.5   | 8.8   | 7.9   | 10.0  | 11.4  | 10.6  |
| (15) Precipitation                                                     | PrecAvg     | 44.9        | 1.6   | 1.2   | 1.2   | 2.0   | 4.8   | 6.3   | 4.4   | 5.8   | 6.9   | 6.2   | 2.7   | 1.7   |
|                                                                        | PrecMax     | 69.5        | 4.9   | 3.8   | 3.5   | 6.3   | 15.9  | 15.4  | 8.0   | 8.4   | 15.8  | 13.3  | 9.2   | 7.4   |
|                                                                        | PrecMin     | 27.6        | 0.5   | 0.2   | 0.5   | 0.7   | 1.1   | 2.3   | 2.3   | 3.5   | 3.2   | 1.7   | 0.0   | 0.2   |
|                                                                        | PrecStd     | 7.6         | 1.1   | 0.8   | 0.6   | 1.2   | 3.3   | 3.0   | 1.4   | 1.2   | 2.4   | 3.2   | 2.0   | 1.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 84.3  | 84.5  | 84.6  | 86.4  | 88.2  | 90.0  | 91.5  | 91.6  | 93.2  | 89.4  | 86.4  | 84.6  |
|                                                                        |             | MCWB        | 74.5  | 74.8  | 74.1  | 76.2  | 77.2  | 78.4  | 79.5  | 79.6  | 82.5  | 78.6  | 77.1  | 76.1  |
|                                                                        | 2%          | DB          | 82.6  | 83.8  | 84.3  | 86.1  | 87.8  | 89.6  | 90.0  | 90.0  | 90.0  | 88.1  | 85.9  | 84.2  |
|                                                                        |             | MCWB        | 73.8  | 74.3  | 74.1  | 76.4  | 77.0  | 78.1  | 78.5  | 78.4  | 78.9  | 78.7  | 77.0  | 75.7  |
|                                                                        | 5%          | DB          | 82.1  | 82.6  | 82.8  | 84.6  | 86.4  | 88.1  | 89.7  | 89.8  | 89.5  | 87.7  | 84.4  | 82.8  |
|                                                                        |             | MCWB        | 73.5  | 73.5  | 73.2  | 75.3  | 76.1  | 78.1  | 78.3  | 78.2  | 78.4  | 78.7  | 76.3  | 74.5  |
|                                                                        | 10%         | DB          | 80.7  | 82.0  | 82.4  | 84.3  | 86.0  | 87.8  | 89.2  | 89.3  | 88.1  | 86.3  | 83.8  | 82.3  |
|                                                                        |             | MCWB        | 72.7  | 73.1  | 72.9  | 75.0  | 75.9  | 78.1  | 78.2  | 78.2  | 78.4  | 77.9  | 75.9  | 74.2  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 76.4  | 76.6  | 77.2  | 78.4  | 79.9  | 81.6  | 81.9  | 81.8  | 82.5  | 81.0  | 79.1  | 78.3  |
|                                                                        |             | MCDB        | 80.9  | 82.1  | 82.8  | 84.8  | 85.1  | 86.2  | 85.5  | 86.8  | 90.5  | 85.4  | 83.1  | 81.9  |
|                                                                        | 2%          | WB          | 75.3  | 75.6  | 75.9  | 77.5  | 79.0  | 80.5  | 80.8  | 80.8  | 81.0  | 80.3  | 78.2  | 77.1  |
|                                                                        |             | MCDB        | 79.7  | 80.8  | 81.6  | 83.5  | 84.1  | 85.8  | 86.5  | 87.0  | 87.3  | 85.3  | 82.8  | 81.4  |
|                                                                        | 5%          | WB          | 74.5  | 74.7  | 75.0  | 76.9  | 78.2  | 79.9  | 80.1  | 80.1  | 80.1  | 79.8  | 77.5  | 76.2  |
|                                                                        |             | MCDB        | 79.3  | 80.0  | 81.0  | 82.7  | 83.5  | 85.4  | 86.4  | 86.6  | 86.2  | 84.9  | 82.5  | 80.8  |
|                                                                        | 10%         | WB          | 73.8  | 74.1  | 74.2  | 76.3  | 77.4  | 79.4  | 79.7  | 79.7  | 79.7  | 79.2  | 76.9  | 75.5  |
|                                                                        |             | MCDB        | 78.9  | 79.5  | 80.3  | 82.1  | 82.9  | 85.0  | 86.1  | 86.3  | 85.9  | 84.5  | 82.3  | 80.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.1   | 8.4   | 8.7   | 8.2   | 8.0   | 7.9   | 8.4   | 8.6   | 8.5   | 7.4   | 7.0   | 7.3   |
|                                                                        |             | MCDBR       | 8.7   | 8.9   | 9.1   | 8.6   | 8.5   | 8.4   | 9.0   | 8.9   | 8.9   | 8.1   | 7.4   | 7.6   |
|                                                                        | 5% WB       | MCWBR       | 4.6   | 4.3   | 4.4   | 4.0   | 4.2   | 4.1   | 4.8   | 4.5   | 4.1   | 4.0   | 4.0   | 4.0   |
|                                                                        |             | MCWBR       | 8.3   | 8.6   | 8.7   | 8.5   | 8.2   | 7.9   | 8.4   | 8.6   | 8.3   | 7.8   | 6.9   | 7.7   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.366 | 0.371 | 0.378 | 0.409 | 0.437 | 0.467 | 0.474 | 0.445 | 0.418 | 0.401 | 0.379 | 0.376 |
|                                                                        |             | taud        | 2.569 | 2.552 | 2.521 | 2.420 | 2.397 | 2.343 | 2.313 | 2.422 | 2.497 | 2.545 | 2.555 | 2.566 |
|                                                                        | Ebn at Noon |             | 288   | 293   | 294   | 284   | 273   | 263   | 261   | 270   | 278   | 279   | 281   | 280   |
|                                                                        |             | Edh at Noon | 30    | 32    | 34    | 38    | 38    | 40    | 42    | 37    | 34    | 31    | 30    | 29    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1386  | 1667  | 1947  | 2149  | 2049  | 1933  | 2086  | 1994  | 1774  | 1563  | 1414  | 1294  |
| (45)                                                                   | RadStd      |             | 96    | 63    | 81    | 50    | 212   | 121   | 125   | 80    | 115   | 138   | 75    | 74    |

Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page